

Advanced Data Mining - Introduction

Course Code	Course Title							
116h54C601	Advanced Data mining							
	TH			P	TUT			Total
Teaching Scheme(Hrs.)	03			--	--			03
Credits Assigned	03			--	--			03
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	30	20	50	--	--	--	--	100

Course prerequisites: Data mining

Course Objectives:

1. To analyze various algorithms and techniques to mine complex data beyond conventional record data to mining complex structure and complex data
2. Extract useful knowledge from massive data sources using distributed computing solutions for data intensive applications
3. To develop the basic skills necessary to pursue research in data mining.

Course Outcomes

On completion of the course students will be expected to

1. Describe the fundamental issues and challenges of mining complex data
2. Analyze patterns in streaming data
3. Derive patterns from complex structures and sequence data
4. Understand the concepts of information retrieval and web search
5. Analyze patterns in multivariate time series data

Module No.	Unit No.	Details	Hrs.	CO
Advanced Data Mining				
1	Data mining Introduction		03	CO1
	Data mining process, different types of data representation, different types of knowledge mined, common data mining tasks, Distributed computing solution for data mining and applications			
2	Incremental Datamining and Stream mining		06	CO2
	Incremental algorithms for mining frequent patterns Characteristics of Streaming Data, Issues and Challenges, Streaming Data Mining Algorithms			
3	Mining complex structures		10	CO3
	Mining trees- Tree Model Guided Framework, TMG framework for mining ordered & unordered subtrees, Tree Mining Applications, Mining maximal and closed frequent trees, Tree mining application Mining Graphs- Approaches to graph mining. Mining social-network graph			
4	Text mining & Web Search		08	CO4
	Text Classification, Vector Space Model, Flat and Hierarchical Clustering Web search: Crawling & Indexing, Hyperlink Analysis, Page Rank algorithm, Web Search and Information Retrieval, Application: Query Recommender System			
5	Sequence mining & Multivariate and Time series mining		08	CO5
	Sequence Mining- Characteristics of Sequence Data, Problem Modeling, Sequential Pattern Discovery, Timing Constraints Applications in Bioinformatics Multivariate and Time series mining- Importance of Multivariate and Time series data, Sources of MVTS data, Mining MVTS data: Sign Language Data, Agro-meteorological Data			

Somaiya Vidyavihar University
K. J. Somaiya College of Engineering, Mumbai -77
(A Constituent College of Somaiya Vidyavihar University)

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Hadzic F., Tan H. & Dillon T. S	Mining data with Complex Structures	Springer	2011
2.	Yates R. B. and Neto B. R	Modern Information Retrieval	Pearson Education	2005
3.	Han J. & Kamber M	Data Mining: Concepts and Techniques	Morgan Kaufmann Publishers	Third edition, 2011
4.	Christopher D.M., Prabhakar R. & Hinrich S	Introduction to Information Retrieval”	Cambridge UP	Online edition, 2009
5.	Tan P. N., Steinbach M & Kumar V	Introduction to Data Mining	Pearson Education	2006

Syllabus

Introduction

What is data?

Introduction

What is data?

Fact or values

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What is Information?

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What is data?

Fact or values

What is Information?

Processed output of data

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What is Information?

Processed output of data

What is Knowledge?

Introduction

What is data?

Fact or values

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Processed output of data

What is Knowledge?

Understanding of information.

X1	X2	X3		Y
2	3	4		1
8	7	5		10
9	6	1		14
7	4	9		2
1	5	5		1
5	3	6		2
7	4	4		7
6	3	5		4
3	2	3		

The Knowledge

What is **data-mining**?

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Computation to facilitate Knowledge Discovery in Databases (**KDD**)

The Knowledge

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Computation to facilitate Knowledge Discovery in Databases (**KDD**)

Goal of **Data Mining** (motivation of doing the same?)

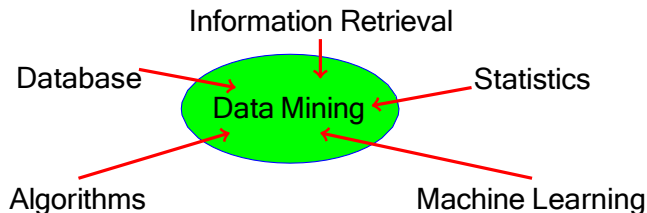
To provide efficient tools and techniques for KDD

Knowledge Discovering in Databases (KDD) involves

- 1 **Selection:** collection of data
- 2 **Preprocessing:** deal with incorrect/missing data
- 3 **Transformation:** common format and preprocessing
- 4 **Data Science: algorithmic tools**
- 5 **Interpretation/Evaluation:** presentation and visualization

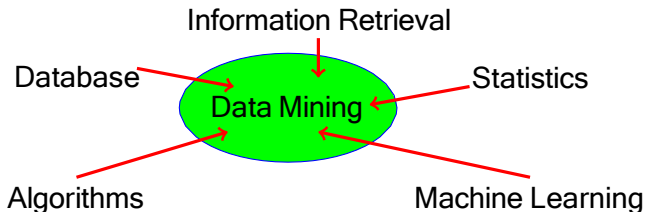
Data Mining

Data mining is fairly involved discipline. It includes many fields such as database, information retrieval, statistics, and machine learning.



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It differs from traditional query processing

- **Query:** not well formed. Miner may not know what he wants.
- **Data:** different version. Preprocessed and modified.
- **Output:** may not a subset. It could be an analysis.

Data Mining Parts

Data Mining has three parts

- 1 **Model:** is to be fit on data
- 2 **Search:** technique to evaluate data point
- 3 **Preference:** criteria to select one model over other

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- 2 Ask for more information
- 3 Decline

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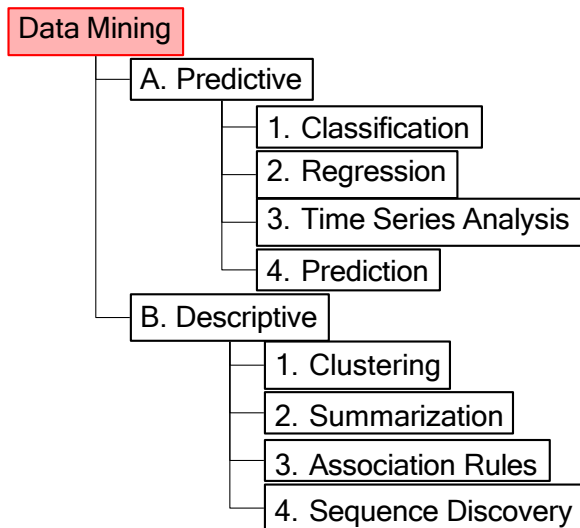
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Data Mining: Tasks

Two broad categories of data mining models are *Predictive* and *Descriptive*. Some of the related tasks are



Classification

Classification maps data into *predefined* labels.



Example: Lots of mails are there in my mail box. Can you tell me which are SPAM?

- Task of supervised learning
- Often based on some patterns or characteristics
- We can use the frequency of words
- Assumption is that some words appears more or less frequently in SPAM

Regression

Regression is used to map data into *real valued* variable.



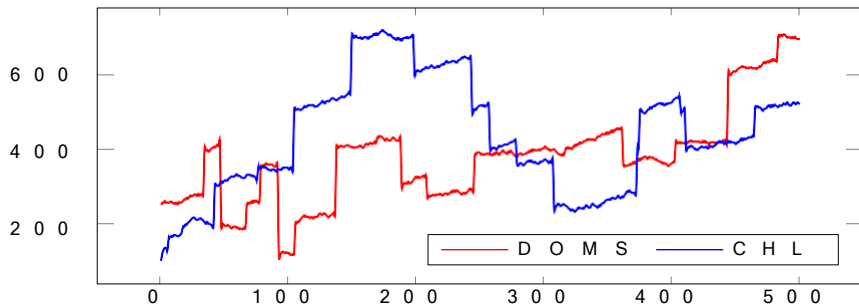
Example: What is the cost of my house?

- Task of supervised learning
- We have data about the cost of house based on features such as
 - › location
 - › Plot area
 - › number of rooms
 - › garden available or not
 - › how old it is
- Current economical conditions can also matter
- Dimensionality is high

Time Series Analysis

In time series analysis the value of attribute is examined over time.

Example: Which stock is better?



- The values are obtained as evenly spaced time points (daily, weekly, hourly, etc.)
- Distance measures are used to find similarity
- Structural analysis is done

Prediction

Predicting future data states based on current or historical data.



Example: What comes next?

2, 4, 6, 8, 10, ...?...

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- Predication can sometimes be seen as classification
- Application includes weather, flood, pattern recognition.

Clustering

Clustering is similar to classification except the groups are not pre-defined.



Example: How many kind of files are there in my directory?

- Unsupervised learning setting
- We can use file name
- Words it has

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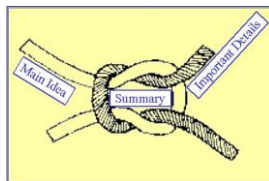
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- Words it has

Example: Who would take my offer?

- The database has information about age, gender, income, location, .. etc.

Summarization

Summarization maps data into subsets with associated simple descriptions. It is also called characterization or generalization.



Example: How to compare two universities?

- Average JEE rank
- Average number of publication
- Student/Faculty ratio
- Combination

Association Rules

Association rules tries to do linked analysis.

Example: Whether same products are selling together?

- $I = \{i_1, i_2, i_3, \dots, i_m\}$, $T = \{t_1, t_2, t_3, \dots, t_n\}$ and $t_i \subseteq I$
- Minimum support count should be maintained
- Can you see: Subset of frequent items is also frequent
- Apriori analysis

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- Minimum support count should be maintained
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Let's do it:

$t_1 = (1, 3, 4)$, $t_2 = (2, 3, 5)$, $t_3 = (1, 2, 3, 5)$, $t_4 = (2, 5)$, $t_5 = (1, 3, 5)$ and minimum support count be 2

Association Rules

$t_1 = (1, 3, 4)$, $t_2 = (2, 3, 5)$, $t_3 = (1, 2, 3, 5)$, $t_4 = (2, 5)$, $t_5 = (1, 3, 5)$

Symb

{1}

{2}

{3}

{4}

{5}

Association Rules

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Symb	Sup
<u>{1}</u>	3 C
<u>{2}</u>	3 C
<u>{3}</u>	4 C
<u>{4}</u>	1
<u>{5}</u>	4 C

Association Rules

$t_1 = (1, 3, 4)$, $t_2 = (2, 3, 5)$, $t_3 = (1, 2, 3, 5)$, $t_4 = (2, 5)$, $t_5 = (1, 3, 5)$

Symb	Sup
{1}	3 C
{2}	3 C
{3}	4 C
{4}	1
{5}	4 C

Symb	Sup
{1,2}	1
{1,3}	3 C
{1,5}	2 C
{2,3}	2 C
{2,5}	3 C
{3,5}	3 C

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Symb	Sup
{1,2}	1
{1,3}	3 C
{1,5}	2 C
{2,3}	2 C
{2,5}	3 C
{3,5}	3 C

Symb	Sup
{1,2,3}	1
{1,2,5}	1
{1,3,5}	2 C
{2,3,5}	2 C

Symb	Sup
{1,2,3,5}	1

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{1,5}	2 C
{2,3}	2 C
{2,5}	3 C
{3,5}	3 C

Symb	Sup
{1,2,3}	1
{1,2,5}	1
{1,3,5}	2 C
{2,3,5}	2 C

Symb	Sup
{1,2,3,5}	1

Try yourself:

Let $I = \{A, B, C, D, E, F\}$ and $T = \{ t_1 = (A, B, C), t_2 = (A, F), t_3 = (A, B, C, E), t_4 = (A, B, D, F), t_5 = (C, F), t_6 = (A, B, C), t_7 = (A, B, C, E), t_8 = (C, D, E), t_9 = (B, D, E) \}$ and min support 3

Sequence Discovery

Sequence Discovery is used to discover sequential patterns in the data.

Example: what is my website access pattern?

- Pattern is based on a time sequence of a action
- It is pattern discovery problem

KDD Issues

- Human interaction
- Overfitting, Outliers
- Large dataset
- High dimension
- Multimedia data
- Missing data
- Irrelevant data
- Noisy data
- Changing data



And much more...

Thank You!

Thank you very much for your attention!

Queries ?